Lesson Objective

Decompose fractions using area models to show equivalence

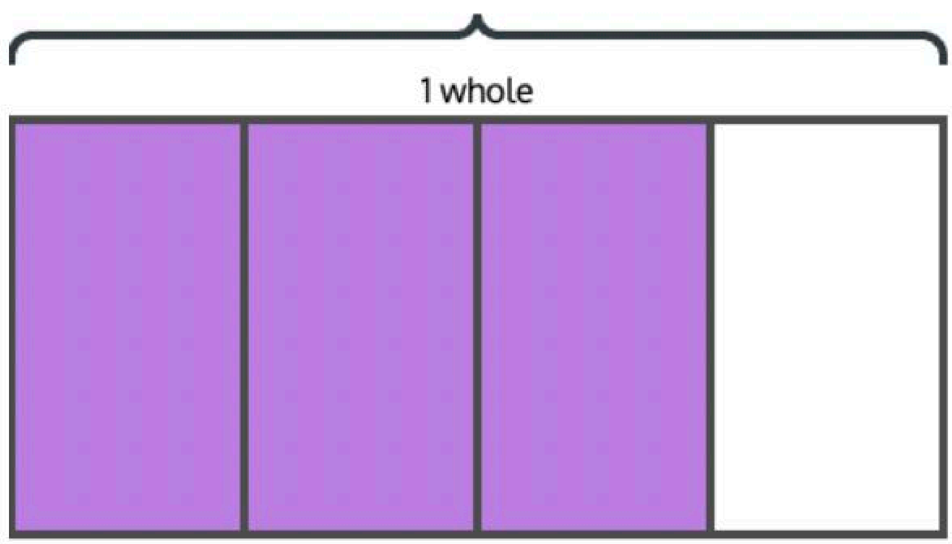
Materials

(S) Personal white board, Mini-Lesson Worksheet

PROBLEM

1 **Use an area model to decompose fourths into eighths, and write addition and multiplication sentences to show the equivalence.**

The rectangle below represents 1 whole and is shaded to show 3/4.



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Draw a horizontal line to decompose the fourths into eighths. Then, write an addition sentence using parentheses and a multiplication sentence to show that 3/4 = 6/8.

Answer: (1/8 + 1/8) + (1/8 + 1/8) + (1/8 + 1/8) = 6/8; 6 × 1/8 = 6/8; 3/4 = 6/8

TEACHER GUIDANCE

Tell students that the rectangle represents 1 whole with 3/4 shaded.

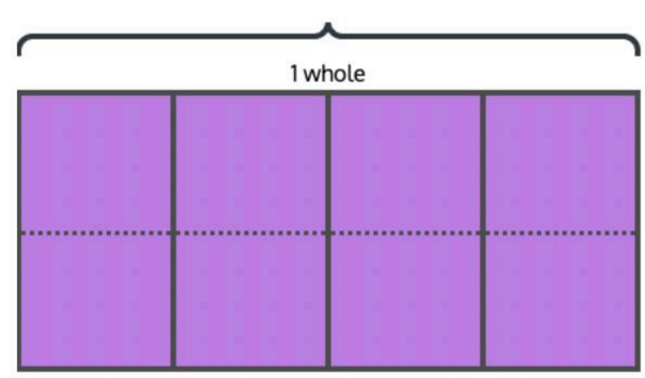
Ask students how many fourths are shaded and how they could split each fourth into 2 equal parts.

Model drawing one horizontal line across the rectangle to create eighths. Ask students how many eighths are shaded now.

Guide students through writing an addition sentence, using parentheses to group the 2 eighths that make up each fourth: (1/8 + 1/8) + (1/8 + 1/8) + (1/8 + 1/8) = 6/8.

Guide students through writing a multiplication sentence: 6 × 1/8 = 6/8.

Point out that the shaded area did not change — the number of pieces increased and each piece got smaller, but the total amount stayed the same, so 3/4 = 6/8.



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PROBLEM	TEACHER GUIDANCE

2 **Draw an area model to show that two-thirds is equivalent to eight-twelfths, and express the equivalence using addition and multiplication of unit fractions.**

Draw an area model to show that 2/3 = 8/12. Write an addition sentence using parentheses and a multiplication sentence to prove the equivalence.

Find the slope of line *a*.

Answer: (1/12 + 1/12 + 1/12 + 1/12) + (1/12 + 1/12 + 1/12 + 1/12) = 8/12; 8 × 1/12 = 8/12; 2/3 = 8/12

Ask students which fraction to draw first and why. Have students draw a rectangle representing 1 whole, partition it into thirds, and shade 2/3.

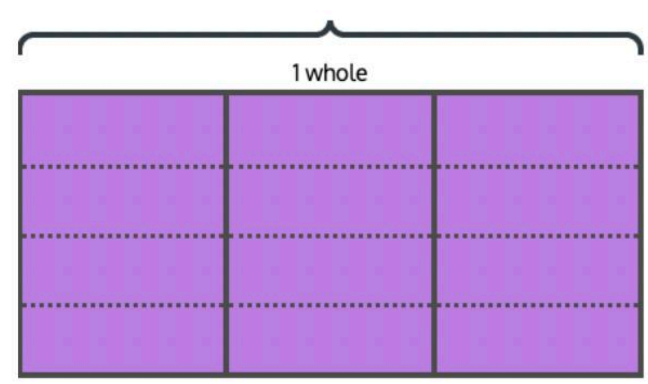
Ask students how they can decompose thirds into twelfths. Guide students to see that each third must be split into 4 equal parts, so they draw 3 horizontal lines across the model.

Ask students how many total parts are in the whole now and how many are shaded.

Have students write an addition sentence using parentheses to group the 4 twelfths within each third: (1/12 + 1/12 + 1/12 + 1/12) + (1/12 + 1/12 + 1/12 + 1/12) = 8/12.

Have students write a multiplication sentence: 8 × 1/12 = 8/12.

Ask students what these sentences tell them about 2/3 and 8/12. Confirm that the shaded area is the same — 2/3 and 8/12 are equivalent fractions.



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3 **Draw an area model to represent a fraction greater than one, then partition it into smaller units to find an equivalent fraction.**

Draw an area model to represent 5/3. Then, partition the thirds into sixths to find an equivalent fraction.

Answer: 5/3 = 10/6

Students work independently.

Monitor For

- Do students draw the horizontal line(s) across the entire rectangle, creating equal-sized parts in every column, not just in the shaded region?
- Do students correctly count the total number of smaller parts in the whole to determine the new denominator (e.g., 8 parts total means eighths, not just counting shaded parts)?
- Do students use parentheses to group the unit fractions that compose each original unit (e.g., grouping the 2 eighths within each fourth)?
- In Problem 3, do students extend the area model beyond 1 whole to represent 5/3, drawing a second rectangle for the remaining 2 thirds?

Instructional Moves

- If a student only draws lines in the shaded region, ask how many equal parts must be in the whole and have them extend the lines across the entire rectangle.
- Before students write parentheses, have them circle the smaller unit fractions on the area model that came from one original unit — this connects the grouping to what they see.
- Remind students that 5/3 means 5 copies of 1/3 — draw 1 whole (3 thirds) and then 2 more thirds in a second rectangle to represent all 5 thirds.
- If a student writes 6 × 1/8 but cannot explain why, ask them to count the shaded eighths one by one and connect each count to the addition sentence.

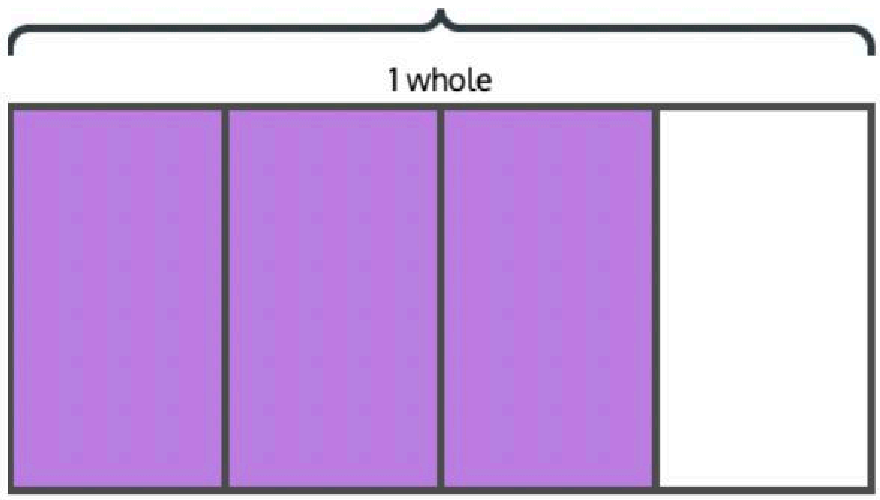
Reflection Questions

1. When we drew a horizontal line to split each fourth into eighths, what happened to the number of parts? What happened to the size of each part?
   Answer: The number of parts doubled — there were more parts, but each part was smaller. The total shaded area stayed the same.
2. Why did we use parentheses when writing the addition sentence for 3/4 = 6/8?
   Answer: The parentheses group the eighths that make up each fourth, so we can see that each 1/4 is the same as 2 copies of 1/8.
3. How did you represent 5/3 with an area model, and how did you know it would need more than 1 whole?
   Answer: 5/3 means 5 copies of 1/3, but 1 whole only has 3 thirds, so I needed a second rectangle for the extra 2 thirds.

Mini Lesson Student Materials

Name ______________________________________

1. The table below shows Biscuit the dog's weight at different ages.



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Draw a horizontal line to decompose the fourths into eighths. Then, write an addition sentence using parentheses and a multiplication sentence to show that 3/4 = 6/8.

2. Draw an area model to show that 2/3 = 8/12. Write an addition sentence using parentheses and a multiplication sentence to prove the equivalence.

3. Draw an area model to represent 5/3. Then, partition the thirds into sixths to find an equivalent fraction.

Mini Lesson Sample Script

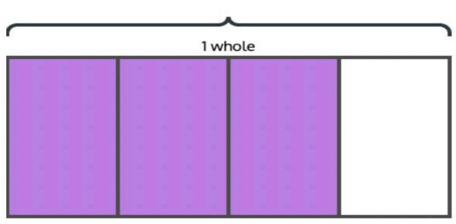
Materials

(S) Personal white board, Mini-Lesson Worksheet

PROBLEM	SAMPLE SCRIPT

1 **Use an area model to decompose fourths into eighths, and write addition and multiplication sentences to show the equivalence.**

The rectangle below represents 1 whole and is shaded to show 3/4.



{"type": "fractionAreaModel", "modelType": "rectangle", "horizSegments": 4, "sectionsShaded": 3, "shadedColor": "purple", "hasWholeLabel": true}

Draw a horizontal line to decompose the fourths into eighths. Then, write an addition sentence using parentheses and a multiplication sentence to show that 3/4 = 6/8.

Answer: (1/8 + 1/8) + (1/8 + 1/8) + (1/8 + 1/8) = 6/8; 6 × 1/8 = 6/8; 3/4 = 6/8

T: Look at the rectangle on your worksheet. It represents 1 whole. How many equal parts is it divided into?

S: 4 parts.

T: And how many are shaded?

S: 3 of them. So 3/4 is shaded.

T: Right. Now, I want to decompose each fourth into 2 equal parts to make eighths. I'm going to draw one horizontal line straight across the middle of the rectangle. Go ahead and do that on yours.

S: (Draw.)

T: How many equal parts are in the whole rectangle now?

S: 8 parts.

T: So each part is what size?

S: 1/8.

T: How many eighths are shaded?

S: 6 eighths.

T: So 3/4 is the same as 6/8. The shaded area didn't change — we just have more pieces, and each piece is smaller. Now let's write an addition sentence. Each fourth became 2 eighths, so we can group them with parentheses: (1/8 + 1/8) for the first fourth, plus (1/8 + 1/8) for the second, plus (1/8 + 1/8) for the third. Write that on your board.

S: (1/8 + 1/8) + (1/8 + 1/8) + (1/8 + 1/8) = 6/8.

T: Good. Now, how many eighths is that in all?

S: 6 eighths.

T: So we can write a multiplication sentence too. How many copies of 1/8?

S: 6 copies. 6 × 1/8 = 6/8.

T: And that tells us 3/4 equals 6/8.

PROBLEM	SAMPLE SCRIPT
2 **Draw an area model to show that two-thirds is equivalent to eight-twelfths, and express the equivalence using addition and multiplication of unit fractions.** Draw an area model to show that 2/3 = 8/12. Write an addition sentence using parentheses and a multiplication sentence to prove the equivalence. *Answer: (1/12 + 1/12 + 1/12 + 1/12) + (1/12 + 1/12 + 1/12 + 1/12) = 8/12; 8 × 1/12 = 8/12; 2/3 = 8/12*	**T**: This time, you're going to draw the area model. We want to show that 2/3 = 8/12. Which fraction should you draw first? **S**: 2/3, because thirds are the bigger pieces. **T**: Go ahead. Draw a rectangle for 1 whole, divide it into thirds, and shade 2 of them. **S**: (Draw and shade.) **T**: Now, we need twelfths. If there are already 3 columns for thirds, how many rows do you need to make 12 equal parts? **S**: 4 rows. Because 3 times 4 is 12. **T**: So how many horizontal lines do you need to draw? **S**: 3 horizontal lines to make 4 rows. **T**: Draw them across the whole rectangle. **S**: (Draw.) **T**: How many total parts are in the whole now? **S**: 12. **T**: And how many are shaded? **S**: 8. **T**: So 2/3 equals 8/12. Now write the addition sentence. Use parentheses to group the twelfths inside each third. **S**: (1/12 + 1/12 + 1/12 + 1/12) + (1/12 + 1/12 + 1/12 + 1/12) = 8/12. **T**: And the multiplication sentence? **S**: 8 × 1/12 = 8/12. **T**: What do those sentences tell you about 2/3 and 8/12? **S**: They're equivalent. The shaded area is the same amount — just described with smaller pieces.
3 **Draw an area model to represent a fraction greater than one, then partition it into smaller units to find an equivalent fraction.** Draw an area model to	**T**: Now try this one on your own. Draw an area model for 5/3, then partition the thirds into sixths to find an equivalent fraction. **S**: (Work.)

PROBLEM	SAMPLE SCRIPT
represent 5/3. Then, partition the thirds into sixths to find an equivalent fraction. *Answer: 5/3 = 10/6*	